Mapping Class Group Orbits of Closed Curves on the Punctured Torus: Computed Counts and Tests of Conjectures

Contents

1 Setup and Conventions **1**
1.1 Curves, Words, Orbits . . . 1

2 Algorithms **2**
2.1 Self-Intersection Number . . . 2
2.2 Orbits via Whitehead's Algorithm . . . 2
2.3 Exhaustive Search . . . 2
2.4 Checks . . . 3

3 Orbit Counts **3**
3.1 Main Table . . . 3
3.2 The Case $k = 4$ . . . 4

4 Split by Filled Subsurface **6**

5 Growth of $N(k)$ **6**
5.1 Known Bounds . . . 6
5.2 Fits to the Data . . . 7

6 Chas's Conjectures on Orbit Sizes **7**
6.1 Statements . . . 8
6.2 Method . . . 8
6.3 Results . . . 8

7 From Level $k-1$ to Level k **9**
7.1 Smoothing a Crossing . . . 9
7.2 Loop Insertion and the Stable Range . . . 10

8 Open Questions **10**

1 Setup and Conventions

1.1 Curves, Words, Orbits

Let T be the once-punctured torus, with $\pi_1(T) \cong F(a,b)$ free on two generators and $A := a^{-1}$, $B := b^{-1}$. Free homotopy classes of oriented closed curves on T are conjugacy classes of $F(a,b)$, i.e. cyclically reduced words up to rotation (*cyclic words*). The puncture is the class of abAB.

Definition 1 (Self-Intersection Number)**.** For a free homotopy class w, $\mathrm{SI}(w)$ is the least number of transverse double points among general-position curves in the class.

Definition 2 (Orbit)**.** By Dehn-Nielsen-Baer, the extended mapping class group of T (all homeomorphisms up to isotopy, orientation-reversing included) is $\mathrm{Out}(F(a,b)) \cong \mathrm{GL}_2(\mathbb{Z})$. An *orbit* is an orbit of this group on the set of *unoriented* classes, so w and w^{-1} are the same curve. Let $N(k)$ be the number of orbits with $\mathrm{SI} = k$.

Definition 3 (Primitive)**.** A class is *primitive* if it is not a proper power v^n, $n \geq 2$. This is not the free-group sense of belonging to a basis. Let $N_{\mathrm{prim}}(k)$ be the number of primitive orbits with $\mathrm{SI} = k$.

Fact 4 (Powers) — For v primitive and $n \geq 1$,

$$\mathrm{SI}(v^n) = n^2\,\mathrm{SI}(v) + n - 1\,.$$

Hence the non-primitive orbits at level k are determined by the primitive orbits at lower levels. For $k = 4$ they are a^5 and $(\mathrm{abAB})^5$; for $k = 5$ they are a^6, $(\mathrm{abAB})^6$, $(\mathrm{aabAB})^2$, $(\mathrm{abaB})^2$.

Remark 1 (other conventions)**.** Three variants were also computed: oriented curves; orientation-preserving mapping classes only (the index-two subgroup of determinant $+1$); both. The counts differ from $k = 3$ on, since some classes are not equivalent to their inverse, resp. mirror image. See Table 2.

2 Algorithms

All code is in `src/`. The original Java pipeline is in `old/`.

2.1 Self-Intersection Number

Proposition 5 (Linked Pairs) Embed the Cayley tree of $F(a, b)$ in the universal cover of T, with the four edges at each vertex in counterclockwise order a, b, A, B. The lifts of a primitive class w are the axes of its conjugates. Two axes cross iff they meet in a (possibly degenerate) segment and arrive at it on opposite sides from how they leave it. Then $\mathrm{SI}(w)$ is the number of crossing pairs of axes up to the action of $F(a, b)$.

Implementation. For each pair of positions (i, j) in the cyclic word, and each of the two relative orientations, walk along the common segment starting at the base vertex and compare the cyclic order of the incoming and outgoing letters. Each unordered pair of axes is found twice. Cost is $O(L^2)$ walks for a word of length L. This is `freegroup.self_intersection`; the same idea with two words gives `freegroup.intersection`.

2.2 Orbits via Whitehead's Algorithm

Theorem 6 (Whitehead, peak reduction) Let W be the finite set of Whitehead automorphisms of $F(a, b)$.

(a) If a cyclic word is not of least length in its Aut-orbit, some element of W strictly shortens it.

(b) Two least-length cyclic words in the same orbit are joined by a chain of elements of W passing only through least-length words.

(c) More generally every word of the orbit is reached from a least-length word by a chain along which length never decreases.

A cyclic word of least length in its orbit is called *minimal*. The *minimal set* of an orbit is its finite set of minimal words.

Algorithm 1 (Canonical orbit representative) **—** Shorten w greedily to a minimal word, search the minimal set breadth-first under W (adding $w \mapsto w^{-1}$ for unoriented curves), and return its lexicographically least element. Tracking the determinant along the search gives the orbit under orientation-preserving classes only.

2.3 Exhaustive Search

Proposition 7 (Length Bound) Every orbit with $\mathrm{SI} = k$ contains a word of length $\leq 4(k + 1)$.

Proof. By the basepoint lemma of Crisp et al. (1998), a curve with k self-intersections is freely homotopic to a product of $k + 1$ simple loops meeting only at one point. By the loop classification of Butler et al. (1995), in a suitable basis each of these is one of a, b, ab, abA, baB or a rotation of abAB. Each has length ≤ 4. ∎

Lemma 8 (Forced Crossings) Whether two lifts cross is decided by the letters along their common segment and one further letter at each end. For a linear word x let $f(x)$ be the number of crossing pairs of lifts decided inside x. Then f is non-decreasing along prefixes, and $f(x) \le \mathrm{SI}(w)$ for every cyclic word w containing x.

Algorithm 2 (`bruteforce.py`) — Depth-first search over prefixes of cyclic words in canonical rotation (prenecklaces), of length $\le 4(k+1)$, abandoning a prefix once $f > k$. Keep the minimal words with $\mathrm{SI} = k$ and sort them into orbits. By Prop. 7 this finds every orbit.

Remark 2. The search uses neither hand-drawn basepoint diagrams, nor loop arrangements, nor the unproven claim that one basepoint diagram suffices for $k = 4$. Running it to lengths $22, 24, 26$ for $k = 2, 3, 4$ finds no orbit beyond those of Prop. 7.

2.4 Checks

- `self_intersection` agrees with the Java library of Chas on $29{,}518$ words: every class of length ≤ 10, and $20{,}000$ random words of length ≤ 40.
- Orbit representatives are invariant under random automorphisms and inversion.
- The pruned search equals a naive enumeration of all words of length ≤ 11, for $k \le 4$.
- $N_{\mathrm{prim}}(k) = 2, 2, 6, 14$ for $k = 0, \dots, 3$ agrees with Crisp et al. (1998); Chas (2016), and the 14 orbits at $k = 3$ are exactly those of Table 2 of Chas (2016).
- Minimal-position drawings (`draw.py`, geodesics on a one-holed torus) have exactly $\mathrm{SI}(w)$ crossings for every orbit with $k \le 5$ and 300 random words.

3 Orbit Counts

3.1 Main Table

k	$N(k)$	$N_{\mathrm{prim}}(k)$	annular	pants	filling
0	2	2	2	0	0
1	4	2	2	2	0
2	8	6	2	5	1
3	16	14	2	11	3
4	29	27	2	18	9
5	57	53	2	34	21
6	109	107	2	60	47
7	204	202	2	110	92
8	369	367	2	181	186
9	669	661	2	306	361
10	1179	1177	2	502	675

Table 1: Orbits of unoriented curves under all homeomorphisms, by exhaustive search. The last three columns split $N(k)$ as in Sec. 4.

k	unoriented, all	oriented, all	unoriented, or.-pres.	oriented, or.-pres.
0	2	2	2	3
1	4	4	4	6
2	8	8	8	14
3	16	17	17	30
4	29	35	35	66
5	57	73	74	139
6	109	149	152	292
7	204	309	310	604
8	369	596	603	1183
9	669	1127	1134	2235
10	1179	2058	2074	4102

Table 2: The same count under the four conventions (powers included).

Remark 3. At $k = 3$ exactly one orbit, that of aabAbaBAb, is not equivalent to its inverse. At $k = 4$ there are six. **[[** possibly the source of the count 15 in Butler et al. (1995); not checked against their list **]]**

Remark 4 (OEIS)**.** None of the sequences in Tables 1 and 2, their partial sums, or their first differences appears in the OEIS, even restricted to runs of five consecutive terms. None satisfies a linear recurrence of order ≤ 4. No orbit counts for $k \geq 4$ were found in the literature.

3.2 The Case $k = 4$

The 29 orbits, one shortest word each:

aaaaa, aaabbb, aabbaB, aaaaabb, aaaabaB, aaabaaB, aaaaabAB, aaaabAAB, aaabAAAB, aabaBAbb, aaabABBAb, aabaBAbAB, aaaabbaBAb, aaabaBabAB, aaabaBAbaB, aabaaBabAB, aabbABabAB, aabABabbAB, aaaabABabAB, aaabAABabAB, aabABabABBAb, aaabbaBAbaBAb, aabaBabABabAB, aabaBAbaBAbaB, aaabABabABabAB, aabAABabABabAB, aabbaBAbaBAbaBAb, aabABabABabABabAB, abABabABabABabABabAB

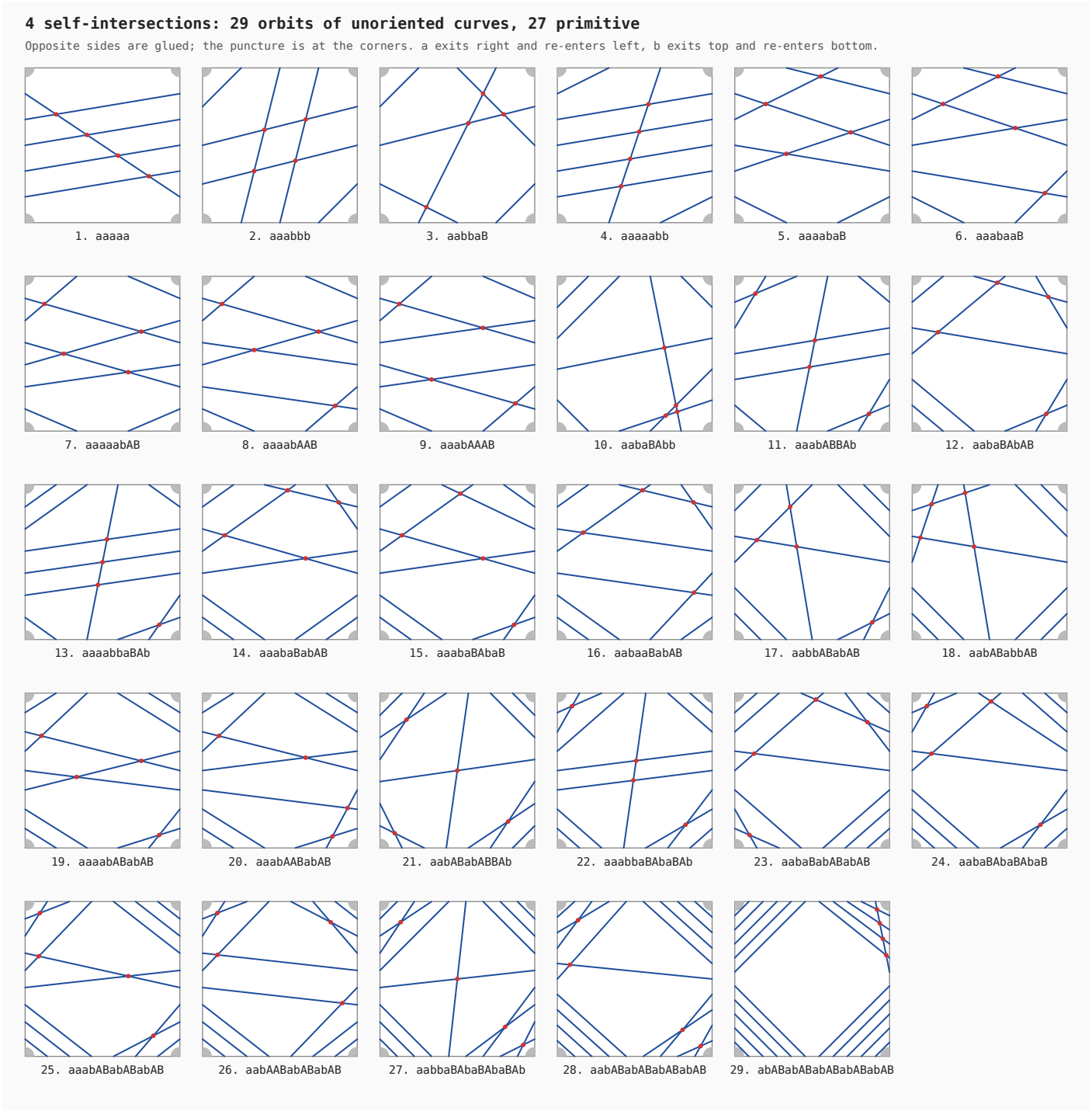


Figure 1: One curve from each orbit with four self-intersections, in minimal position. Opposite sides of each square are glued and the puncture is at the corners; a exits right, b exits top.

Earlier claims.

- `Punctured Torus Count.tex` lists 44 words; these contain only 28 distinct orbits, and the orbit of aabbaBAbaBAbaBAb is missing.
- The Java output of 33 words covers all 29 orbits, counting four of them twice as a word and its inverse; as an oriented count the truth is 35 and two are missing.
- The hand-made loop arrangements for $k = 3$ never produce the orbit of abaBAbAB; the Java total of 16 matched the

literature only because one orbit and its inverse were counted separately.

- The arrangement files contain a malformed ordering (nine entries), after which the Java reader silently drops the rest of the file, and four arrangements that produce curves with more than k self-intersections, which no k-crossing picture can.

4 Split by Filled Subsurface

Definition 9. A class is *annular* if it is a power of a simple curve or of the puncture; *pants* if it is not annular and is disjoint from some essential non-peripheral simple closed curve; *filling* otherwise.

Lemma 10 (a) A pants class is disjoint from exactly one simple closed curve α, and lies in the pair of pants $T \setminus \alpha$.

(b) A class is disjoint from the simple curve a iff it is conjugate into $\langle a, bAB\rangle$, iff the letters b, B alternate around its cyclic word.

(c) Such an alternating word is minimal.

Proof. (a) Two distinct such curves fill T, so a class disjoint from both is peripheral. (b) $\pi_1(T \setminus a) = \langle a, bAB\rangle$. (c) Write the word as $x^{i_1}y^{j_1}\cdots x^{i_m}y^{j_m}$ with $x = a$, $y = bAB$. The Whitehead automorphisms multiplying b by $a^{\pm 1}$ fix the cyclic word. Those multiplying a by $b^{\pm 1}$ change its length by $\sum_r(|i_r| + |j_r|) - 2m \geq 0$. ∎

So an orbit is non-filling iff some word of its minimal set is alternating; this is `filling.py`.

Remark 5 (checks)**.** For all 225 orbits with $k \leq 6$, and all simple curves of word length ≤ 9: pants orbits have zero intersection number with exactly one simple curve, filling orbits with none, powers of the puncture with all. Independently, enumerating words in the pants generators up to the length bound $2k + 2$ of Chas and Phillips (2012) and sorting into orbits reproduces the pants column for $k \leq 5$.

Remark 6. The pants column counts closed curves on a pair of pants with k self-intersections up to the four symmetries fixing one boundary component. Filling orbits overtake pants orbits at $k = 8$; their step ratio is close to 2 ($1.96, 2.02, 1.94, 1.87$ for $k = 7, \ldots, 10$) against $1.83, 1.65, 1.69, 1.64$ for pants.

5 Growth of $N(k)$

5.1 Known Bounds

Let $O(K, S)$ be the number of orbits with $\mathrm{SI} \leq K$ on a surface S.

Theorem 11 (Sapir 2016) For any compact S and $K \geq 1$,

$$\tfrac{1}{12}\, 2^{\sqrt{K/12}} \leq O(K, S) \leq \exp\big(d_S \sqrt{K} \log(d_S \sqrt{K})\big)\,.$$

The lower bound comes from curves in an embedded pair of pants, whose stabilizer in the mapping class group is finite.

Theorem 12 (Aougab and Souto 2016) For S closed and every $\delta > 0$, for K large,

$$\exp\big((\pi\sqrt{|\chi(S)|} - \delta)\sqrt{K}\big) \leq O(K, S) \leq \exp\big((4\sqrt{2|\chi(S)|} + \delta)\sqrt{K}\big)\,.$$

The upper bound holds for every finite-type orientable S with $\chi(S) < 0$ (their Sec. 7.1). They conjecture that the lower bound is sharp.

Claim 13 (research note `curve_orbit_bounds.pdf`, not independently verified) — For $S = T$ and K large, the number of orbits with SI *exactly* K is at least $\frac{c}{\sqrt{K}}\, 3^{3\sqrt{K}-9K^{1/4}}$, so that

$$3\log(3) \le \liminf_{K\to\infty} \frac{\log N(K)}{\sqrt{K}} \le \limsup_{K\to\infty} \frac{\log N(K)}{\sqrt{K}} \le 4\sqrt{2}\,.$$

The construction takes a typical word of length $\approx 3\sqrt{K}$ in a pair of pants, where $\mathrm{SI} \approx N^2/9$ (Chas and Lalley, 2012), and pads its unique longest block.

Lemma 14 (Padding, from the same note) In pants generators, if $a^M u$ is cyclically reduced, u begins and ends with $b^{\pm 1}$, and every $a^{\pm 1}$-block of u is shorter than M, then $\mathrm{SI}(a^{M+d}u) = \mathrm{SI}(a^M u) + d$ for all $d \ge 0$.

Remark 7 (checks). The padding identity holds in $4{,}500$ random cases, and fails in 426 of $1{,}500$ random cases when the unique-longest-block hypothesis is dropped. The transfer from pants to T is the argument of Sapir (2016). The statement for exactly K, and the constant $3\log(3)$, were not found in Sapir (2016); Aougab and Souto (2016), which count $\mathrm{SI} \le K$.

[[$3\log(3) \approx 3.30 > \pi$, the value conjectured by Aougab-Souto at $|\chi| = 1$ (stated for closed S). Their intersection form may count each crossing twice, in which case their constant per crossing is $\pi\sqrt{2} \approx 4.44$ and there is no conflict. Convention not settled from their text.]]

5.2 Fits to the Data

k	1	2	3	4	5	6	7	8	9	10
$N(k)/N(k-1)$	2	2	2	1.81	1.97	1.91	1.87	1.81	1.81	1.76

Least squares on $\log N(k)$, $k = 1, \dots, 8$:

model	fitted	rms error	predicted $N(9)$	predicted $N(10)$
$\exp(c\sqrt{k})$	$c = 2.5$	0.18		
$k^{-a}\exp(c\sqrt{k})$	$c = 4.45$, $a = 1.76$	0.034	≈ 627	≈ 1070
$\exp(\alpha k)$	$e^\alpha = 1.91$	0.032	≈ 736	≈ 1400
$k^a \exp(\alpha k)$	$e^\alpha = 1.84$, $a = 0.14$	0.018	≈ 700	≈ 1310
actual			669	1179

Remark 8.
- Bare $\exp(c\sqrt{k})$ does not fit; a power-law prefactor is needed.
- With the prefactor, c is unstable: 4.45 on $k \in [1, 8]$, 5.5 on $k \in [3, 8]$.
- The data for $k \le 8$ do not separate $\exp(c\sqrt{k})$ from $\exp(\alpha k)$; the values at $k = 9, 10$ fall between the two predictions.
- The asymptotic regime is far away: the inequality used in the claim above needs $K^{1/4} \ge 6 + 3\sqrt{3}$, i.e. $K \gtrsim 1.5 \times 10^4$.
- At $k = 10$ the shortest words of the orbits have lengths 9 to 44 with mode 15, against the typical-word length $3\sqrt{k} \approx 9.5$: the orbits seen are long words with unusually few crossings.

[[heuristic: the true constant should exceed $3\log(3)$. Words with slightly fewer crossings per length than typical are only second-order rarer in entropy, but gain first-order in length for the same K.]]

6 Chas's Conjectures on Orbit Sizes

6.1 Statements

For an orbit α of unoriented curves let $c_\ell(\alpha)$ be the number of its classes of word length ℓ, and let ϕ be Euler's function, extended by 0 off the positive integers.

Conjecture 15 (Chas 2016, Conjecture 4)**.** For every α there are integers j_i and $k_i \geq 1$ with

$$c_\ell(\alpha) = 2\sum_{i=1}^{n} \phi\Big(\frac{\ell + j_i}{k_i}\Big)$$

for ℓ large. Then $\sum_{\ell \leq L} c_\ell(\alpha) \sim \frac{6}{\pi^2}\, p_\alpha L^2$ with $p_\alpha := \sum_i k_i^{-2}$.

Conjecture 16 (Chas 2016, Conjecture 5)**.** For each k, the sum of p_α over the primitive non-peripheral orbits with $\mathrm{SI} = k$ is within 1 of the number of such orbits.

6.2 Method

By part (c) of Whitehead's theorem, a breadth-first search from the minimal set under W, discarding words longer than L, finds every class of the orbit of length $\leq L$. This gives $c_\ell(\alpha)$ exactly for $\ell \leq L$ (`orbit_growth.py`). A formula of the shape of Conj. 15 is then searched for, fewest terms first, with $k_i \leq 12$, shifts that are multiples of 4 tried first. Formulas are fitted on $\ell \leq 60$.

6.3 Results

k	orbits	$\sum_\alpha p_\alpha$	difference
0	1	1	0
1	2	$9/4$	$+0.25$
2	6	$197/36 \approx 5.47$	-0.53
3	14	$2023/144 \approx 14.05$	$+0.05$
4	27	$\geq 103951/3600 \approx 28.88$; estimated ≈ 30.2	$\geq +1.88$; $\approx +3.2$
5	53	$\geq 6691/120 \approx 55.76$; estimated ≈ 62.2	$\geq +2.76$; $\approx +9.2$

Table 3: Test of Conj. 16. For $k = 4, 5$ the lower bound is the exact sum over the orbits with a fitted formula (25 of 27, resp. 47 of 53).

Claim 17 — Conj. 16 fails at $k = 4$ and $k = 5$, granting that the fitted formulas hold for all ℓ.

Remark 9 (evidence)**.**

- For $k \leq 3$ the method recovers every formula of Tables 1 and 2 of Chas (2016), including those marked unproven.
- Every formula fitted on $\ell \leq 60$ predicts c_ℓ exactly for $61 \leq \ell \leq 84$: all 25 at $k = 4$ and all 47 at $k = 5$.
- For the orbits without a formula, p_α is estimated by a two-point extrapolation of $\sum_{\ell \leq L} c_\ell$ at $L = 60, 84$; on the fitted orbits this method gives 28.5 against the exact 28.88.

Remark 10. Table 4 of Chas (2016) gives $2059/144$ at $k = 3$; the entries of her Table 2 sum to $2023/144$, which is the value found here.

orbit	c_ℓ	valid for $\ell \geq$	p_α
aaabbb	no formula found		≈ 0.65
aabbaB	$4\phi(\ell-4)+4\phi(\ell)$	7	4
aaaaabb	no formula found		≈ 0.69
aaaabaB	$4\phi(\ell/5)$	8	$2/25$
aaabaaB	$4\phi(\ell/5)$	8	$2/25$
aaaaabAB	$4\phi((\ell-4)/4)$	8	$1/8$
aaaabAAB	$4\phi((\ell-4)/4)$	8	$1/8$
aaabAAAB	$2\phi((\ell-4)/4)$	8	$1/16$
aabaBAbb	$16\phi((\ell-4)/2)$	9	2
aaabABBAb	$2\phi(\ell-8)+2\phi(\ell-4)$	10	2
aabaBAbAB	$8\phi((\ell-4)/3)$	10	$4/9$
aaaabbaBAb	$8\phi((\ell-8)/2)+8\phi((\ell-4)/2)$	11	2
aaabaBabAB	$8\phi((\ell-4)/4)$	11	$1/4$
aaabaBAbaB	$4\phi((\ell-4)/4)$	11	$1/8$
aabaaBabAB	$8\phi((\ell-4)/4)$	11	$1/4$
aabbABabAB	$8\phi(\ell-8)$	11	4
aabABabbAB	$8\phi(\ell-8)$	11	4
aaaabABabAB	$4\phi((\ell-8)/3)$	11	$2/9$
aaabAABabAB	$8\phi((\ell-8)/3)$	11	$4/9$
aabABabABBAb	$4\phi((\ell-8)/2)$	13	$1/2$
aaabbaBAbaBAb	$4\phi(\ell-12)+4\phi(\ell-8)$	14	4
aabaBabABabAB	$8\phi((\ell-8)/3)$	14	$4/9$
aabaBAbaBAbaB	$4\phi((\ell-8)/3)$	14	$2/9$
aaabABabABabAB	$4\phi((\ell-12)/2)$	14	$1/2$
aabAABabABabAB	$4\phi((\ell-12)/2)$	14	$1/2$
aabbaBAbaBAbaBAb	$4\phi((\ell-12)/2)$	17	$1/2$
aabABabABabABabAB	$4\phi(\ell-16)$	17	2

Table 4: Orbit sizes for the 27 primitive orbits with $k=4$. The table for $k=5$ is in `results/growth_k5.txt`.

Remark 11 (on Conj. 15)**.** No formula was found for a^3b^3, a^5b^2 at $k=4$ and for six orbits at $k=5$ (aaaaaabb, aaabbabb and four with $p_\alpha \approx 1.33$). For a^3b^3 the counts along $\ell \equiv 1 \pmod 6$ are $4,4,4,8,12,12,16,16,20,24,24$ for $\ell = 7,13,\dots,67$, which grows smoothly, unlike totient sums. The estimates $p \approx 0.65$ and $p \approx 1.33$ suggest $2/3$ and $4/3$, which are not sums of k_i^{-2} in any evident way. **[[** whether Conj. 15 fails for these orbits is open; a wider search ($k_i \leq 12$, up to 16 terms) timed out without a result **]]**

7 From Level $k-1$ to Level k

7.1 Smoothing a Crossing

At a self-intersection the cyclic word splits as $w = uv$ into two loops based at the crossing. The smoothing that keeps one component is uv^{-1}, with $\mathrm{SI}(uv^{-1}) \leq \mathrm{SI}(w) - 1$. Conversely, if $p = \alpha\beta$ then every curve obtained from p by adding one crossing is

$$\alpha\,\gamma^{-1}\beta^{-1}\gamma^{-1}$$

for some word γ, the class of an arc joining the two points where p is cut.

Claim 18 (verified for $1 \leq k \leq 8$) **—** Every primitive orbit with $\mathrm{SI} = k$ has a crossing whose smoothing has SI exactly $k-1$. Every orbit with $\mathrm{SI} = k-1$, powers included, arises this way.

Remark 12. The relation is many-to-many: at $k=8$ an orbit has 1 to 7 distinct parents and a parent up to 18 children; about 44% of smoothings drop SI by exactly one. Generating level k from minimal words of level $k-1$ with $|\gamma| \leq 2$ is incomplete: it finds 199 of the 202 primitive orbits at $k=7$. An exact generator would need the arcs γ inside faces of all minimal-position diagrams of the parent.

7.2 Loop Insertion and the Stable Range

A *loop insertion* inserts one of the eight rotations of $(\mathrm{abAB})^{\pm 1}$ into a minimal word at any position, keeping the result if it is cyclically reduced as written, minimal, and has one more self-intersection.

Claim 19 (verified for $2 \le k \le 10$) — The orbits with $\mathrm{SI} = k$ whose minimal words have length $\ge 2k+3$ are exactly the loop insertions of the orbits with $\mathrm{SI} = k-1$ whose minimal words have length $\ge 2k-1$.

k	2	3	4	5	6	7	8	9	10
orbits of length $\ge 2k+3$	3	8	11	20	31	58	85	140	213
parents (level $k-1$, length $\ge 2k-1$)	3	7	11	19	31	57	85	139	213
orbits that are not loop insertions	4	7	11	22	41	66	112	191	
their greatest length	6	8	9	12	14	16	17	20	

Table 5: Each parent has exactly one child, except one parent with two at each odd k. The last two rows use loop insertions from all of level $k-1$.

Remark 13. The threshold $2k+2$ is sharp: at most levels some orbit that is not a loop insertion has minimal length exactly $2k+2$. In the stable range the number of orbits with minimal length ℓ at level k equals the number with length $\ell-4$ at level $k-1$ (for even k; at odd k there is one more). Filling orbits have minimal length up to $4k-3$, pants orbits up to $4k+1$, the puncture power $4k+4$.

Algorithm 3 (`fast.py`) — For $k \ge 2$: exhaustive search to length $2k+2$, union the loop insertions of the level $k-1$ orbits of length $\ge 2k-1$. Every orbit reported is verified individually, so if Claim 19 failed at some k the count would be too small, never too large.

k	6	7	8	9	10	11	12
`bruteforce.py` (seconds, 16 cores)	16	66	216		1676		
`fast.py` (seconds, 16 cores)	1.2	2.5	6.0	15.9	48.8	113	268

Table 6: Running times. For $k \le 10$ the two methods return identical sets of orbits.

k	$N(k)$	$N_{\mathrm{prim}}(k)$	annular	pants	filling	$N(k)/N(k-1)$
11	2081	2077	2	840	1239	1.77
12	3602	3600	2	1359	2241	1.73

Table 7: Counts from `fast.py` beyond the range of the exhaustive search. These are exact if Claim 19 holds at $k = 11, 12$, and lower bounds otherwise.

Remark 14. The non-primitive counts agree with the power formula: 4 at $k = 11$ (a^{12}, $(\mathrm{abAB})^{12}$ and the cubes of the two primitive orbits with $k = 1$) and 2 at $k = 12$.

8 Open Questions

1. Prove Claim 19. A natural route: a minimal word of length $\ge 2k+3$ with $\mathrm{SI} = k$ must contain a loop round the puncture whose removal costs exactly one crossing, in analogy with the bound $L \le 2k+2$ of Chas and Phillips (2012) on the pair of pants.
2. Prove Claim 18.
3. Decide Conj. 15 for a^3b^3 and a^5b^2; find p_α exactly for the orbits of a^pb^q.

4. Is there a corrected form of Conj. 16? The excess $\sum_\alpha p_\alpha - N_{\mathrm{prim}}(k)$ is $0, 0.25, -0.53, 0.05, \approx 3.2, \approx 9.2$.

5. The constant in $\log N(k) \sim c\sqrt{k}$, and whether the limit exists; settle the counting convention in Aougab and Souto (2016).

6. A formula or recursion for the pants column, using the block structure of words in $\langle a, bAB \rangle$ and the intersection formula of Diop and Gaye (2022).

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